

Module 05: Simple Mail Transfer Protocol (SMTP) and Gateways

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

SIMPLE MAIL TRANSFER PROTOCOL (SMTP) & GATEWAYS

Connection Establishment, Mail Transaction, Teardown & Enterprise Security Gateways

Sender Mail Server

MTA Client Engine

Outbox Spool

Queued Outgoing Mail

Pushes Messages

Port 25 / 587 (Submission)

SMTP TRANSACTION 3-PHASE INTERACTION (TCP PORT 25)

```
<-- 220 server.com Simple Mail Transfer Ready
--> HELO client.domain.com (or EHLO)
<-- 250 Hello client.domain.com, pleased to meet you
--> MAIL FROM: <alice@aktu.ac.in>
<-- 250 2.1.0 Sender OK
--> RCPT TO: <bob@aktu.ac.in> | <-- 250 Recipient OK
--> DATA | <-- 354 Start mail input; end with <CRLF>.<CRLF>
[Body text / MIME payload...] | --> . | <-- 250 OK: queued
--> QUIT | <-- 221 2.0.0 server.com Service closing transmission
```

Receiver Mail Server

MTA Server Daemon

User Mailbox

Stores Delivered Mail

Listening on Port 25

Enforces Spam Checks

1. Key SMTP Commands

HELO / EHLO: Client identifies identity • **MAIL FROM:** Specifies envelope sender • **RCPT TO:** Specifies destination mailbox • **DATA:** Initiates message payload transfer • **QUIT:** Gracefully tears down TCP link

2. Mail Relays & DNS MX • **Mail Relays:** Intermediate MTAs forwarding emails across autonomous corporate networks. • **DNS MX Lookup:** Resolves destination domain to highest priority mail host IP. • Multiple MX records provide auto-failover.

3. Email Gateways (AKTU PYQ) • **Purpose:** Filtering, virus scan & routing. • **On-Premises / Appliance Gateways:** Cisco IronPort, Barracuda Spam Gateway, Fortinet FortiMail, Symantec Mail Security. • **Cloud Gateways:** Proofpoint, Mimecast.

1. SMTP Operational Paradigm

SMTP is a reliable, push-based application protocol executing over TCP Port 25 for inter-MTA communication and Port 587 for client submission.

Three Discrete Interactive Phases

Every SMTP transmission sequence proceeds through: **1. Connection Establishment** (HELO/EHLO), **2. Mail Transaction** (MAIL FROM, RCPT TO, DATA), and **3. Teardown** (QUIT).

2. Primary SMTP Commands & Status Replies

Command	Server Reply	Semantic Meaning
HELO client.domain	250 OK	Identifies client host identity to the server
MAIL FROM: <sender>	250 Sender OK	Specifies message envelope origin
RCPT TO: <recipient>	250 Recipient OK	Declares destination mailbox destination
DATA	354 Start mail input	Initiates raw text/MIME message body streaming
QUIT	221 Service closing	Graceful session termination

3. Email Gateways (AKTU PYQ Classification)

- **Appliance / On-Premises:** Cisco IronPort (ESA), Barracuda Email Security, Fortinet FortiMail, Symantec Messaging Gateway.
- **Cloud-Hosted:** Proofpoint Enterprise Protection, Mimecast Cloud Gateway.